MATH 110 — Week 1

Homework 1 — Solutions

HW1 P1(a): sending an inverse image forward. **Idea.** To belong to $f^{-1}(V)$, an input must already have its output in V. Sending those inputs forward therefore selects the elements of V that the function actually attains.

1. Let $y \in f(f^{-1}(V))$. By the image definition there is an $x \in f^{-1}(V)$ with $y = f(x)$. By the inverse-image definition, $f(x) \in V$. Thus $y \in V$, proving the requested inclusion.
2. In fact $f(f^{-1}(V)) = V \cap f(X)$. For the forward inclusion, the preceding step gives $y \in V$ and the equation $y = f(x)$ gives $y \in f(X)$.
3. For the reverse inclusion, let $y \in V \cap f(X)$. Because $y \in f(X)$, choose $x \in X$ with $f(x) = y$. Because $y \in V$, this x lies in $f^{-1}(V)$. Hence $y = f(x) \in f(f^{-1}(V))$.

The intersection formula identifies exactly which outputs are lost: the elements of V outside $f(X)$.

HW1 P1(b): taking the inverse image of an image. **Idea.** Every $u \in U$ supplies an output in $f(U)$. The inverse image selects u and also any other input with the same output. To describe the extra inputs, compare one input outside U with one inside U.

1. If $u \in U$, then $f(u) \in f(U)$ by the image definition. Thus $u \in f^{-1}(f(U))$, proving $U \subseteq f^{-1}(f(U))$.
2. More explicitly,
$$f^{-1}(f(U)) = \{x \in X : \text{there exists } u \in U \text{ with } f(x) = f(u)\}.$$
Indeed, $f(x) \in f(U)$ means exactly that some $u \in U$ has value $f(x)$.
3. An extra element therefore exists exactly when there are $x \in X \setminus U$ and $u \in U$ with $f(x) = f(u)$. This is equivalent to the images $f(X \setminus U)$ and $f(U)$ having a common element.

No inverse function is used: $f^{-1}(f(U))$ is defined even if several inputs have the same output.

HW1 P1(c): strict examples. **Idea.** Use one function with both a missing output and repeated outputs. A missing output can make (a) strict; repeated outputs can make (b) strict.

Let $X = \{a, b, c, d\}$, $Y = \{p, q, r\}$ and

x	a	b	c	d
$f(x)$	p	p	q	q

.

1. Set $V = \{q, r\}$. Only c, d map into V, so $f^{-1}(V) = \{c, d\}$. Their outputs form $\{q\}$, so $f(f^{-1}(V)) = \{q\} \subsetneq V$. The missing element is r.
2. Set $U = \{a, c\}$. Then $f(U) = \{p, q\}$ and every input maps into this set. Thus $f^{-1}(f(U)) = X \supsetneq U$. The extra elements are b, d.

HW1 P1(c): exact equality conditions for a fixed subset. **Idea.** The previous formulas identify the two possible obstructions: an unattained element of V, or a value shared by an input in U and one outside U. Excluding the relevant obstruction is both necessary and sufficient.

1. From part (a), $f(f^{-1}(V)) = V \cap f(X)$. This equals V if and only if $V \subseteq f(X)$: each element of this particular V must be attained.
2. From part (b), equality $f^{-1}(f(U)) = U$ holds if and only if

$$f(U) \cap f(X \setminus U) = \varnothing.$$

   If the intersection has an element $y = f(u) = f(x)$ with $u \in U$, $x \notin U$, then x is an extra inverse-image element. Conversely, any extra inverse-image element supplies such a y.
3. For the function in the strict examples, $V = \{p\}$ gives equality in (a), since $f^{-1}(V) = \{a, b\}$ and its image is $\{p\}$, even though f is not surjective.
4. For that same function, $U = \{a, b\}$ gives equality in (b), since $f(U) = \{p\}$ and $f^{-1}(\{p\}) = \{a, b\}$, even though f is not injective.

These examples show why global surjectivity and injectivity are stronger than the equality conditions for a fixed V or U.

HW1 P1(c): equality for every subset. **Idea.** To force surjectivity, test the whole codomain. To force injectivity, test a singleton input set: equality then prevents another input from having the same value.

1. If f is surjective, $f(X) = Y$, so $V \cap f(X) = V$ for every $V \subseteq Y$. Conversely, equality for every V includes $V = Y$; part (a) then gives $Y = f(X)$, hence surjectivity.
2. If f is injective and $x \in f^{-1}(f(U))$, choose $u \in U$ with $f(x) = f(u)$. Injectivity gives $x = u \in U$. Combined with part (b), this proves equality for every $U \subseteq X$.
3. Conversely, suppose equality holds for every U. If $f(x) = f(u)$, take $U = \{u\}$. Then $x \in f^{-1}(f(\{u\})) = \{u\}$, so $x = u$. This is injectivity.

HW1 P2 (LADR 1A.7). **Idea.** To prove that z is a cube root of 1, compute z^3. Squaring first is useful here: the square turns out to be the conjugate of z, so one more multiplication is real.

1. Set $z = (-1 + \sqrt{3}\, i)/2$. Expand the square, using $i^2 = -1$:

$$z^2 = \frac{(-1)^2 + 2(-1)(\sqrt{3}\, i) + (\sqrt{3}\, i)^2}{4} = \frac{1 - 2\sqrt{3}\, i - 3}{4} = \frac{-1 - \sqrt{3}\, i}{2}.$$

2. Multiply z^2 by z and show the imaginary terms cancel:

$$z^3 = \frac{(-1 - \sqrt{3}\, i)(-1 + \sqrt{3}\, i)}{4} = \frac{1 - \sqrt{3}\, i + \sqrt{3}\, i - 3i^2}{4} = \frac{1 + 3}{4} = 1.$$

3. A cube root of 1 means a number whose cube is 1. The computed equality proves the claim.

HW1 P3: a zero product in a field. **Idea.** The inverse axiom lets us remove a only if $a \neq 0$. Split into the two cases first, then multiply the equation by a^{-1} in the second case.

1. If $a = 0$, one of the required alternatives already holds.
2. Suppose $a \neq 0$. The field axiom provides a^{-1}, with $a^{-1}a = 1$. Thus

$$b = 1b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}0 = 0.$$

   Associativity gives the third equality, the hypothesis gives the fourth, and the field multiplication-by-zero identity $c0 = 0$ proved in the handout gives the last.
3. Thus either $a = 0$ or, in the other case, $b = 0$.

HW1 P4 (LADR 1B.2): zero scalar multiple of a vector. **Idea.** The field still supplies an inverse for a nonzero scalar. Apply that inverse through scalar multiplication of vectors, using $(ab)v = a(bv)$; Notes Proposition 4 identifies a scalar multiple of the zero vector.

1. If $a = 0_F$, the first alternative holds. Suppose instead $a \neq 0_F$, so a^{-1} exists.
2. Apply a^{-1} to $av = 0_V$:
$$v = 1v = (a^{-1}a)v = a^{-1}(av) = a^{-1}0_V = 0_V.$$
Here $1v = v$ and scalar associativity are vector-space axioms; $a^{-1}0_V = 0_V$ is Notes Proposition 4(ii).
3. Thus $a = 0_F$ or $v = 0_V$. The argument does not divide by a vector; only the scalar a is inverted.

HW1 P5 (LADR 1C.4): a prescribed integral. **Idea.** A subspace must contain the zero function, whose integral is zero. If this necessary condition gives $b = 0$, check it is sufficient by calculating the integral of $af + g$.

We cite three elementary-calculus facts: continuous functions on $[0, 1]$ have definite integrals, sums and real scalar multiples of continuous functions are continuous, and integration is linear.

1. Suppose U_b is a subspace. Its zero vector is the function $f(x) = 0$ for every x. Since $\int_0^1 0\,dx = 0$, membership in U_b forces $b = 0$.
2. Conversely, assume $b = 0$. The zero function is continuous and has integral zero, so belongs to U_0.
3. Let $f, g \in U_0$ and $a \in \mathbb{R}$. The function $af + g$ is continuous, and linearity gives
$$\int_0^1 (af(x) + g(x))\,dx = a\int_0^1 f(x)\,dx + \int_0^1 g(x)\,dx = a \cdot 0 + 0 = 0.$$
Thus $af + g \in U_0$.
4. The zero and closure conditions are the subspace test (Notes Definition 5). They prove that U_0 is a subspace, completing both directions.

For a concrete member, $f(x) = x - 1/2$ is continuous and
$$\int_0^1 (x - 1/2)\,dx = \left[\frac{x^2}{2} - \frac{x}{2}\right]_0^1 = \frac{1}{2} - \frac{1}{2} = 0.$$

HW1 P6 (LADR 1C.5): the scalar field matters. **Idea.** The ambient space is a complex vector space, so scalar closure must include multiplication by i. One real vector sent outside $\mathbb{R}^2$ disproves closure.

1. The vector $(1, 0)$ belongs to $\mathbb{R}^2 \subseteq \mathbb{C}^2$, and $i \in \mathbb{C}$ is an allowed scalar.
2. Multiplication gives $i(1, 0) = (i \cdot 1, i \cdot 0) = (i, 0)$. Its first coordinate is not real, so $(i, 0) \notin \mathbb{R}^2$.
3. Thus $\mathbb{R}^2$ is not a subspace of the complex vector space $\mathbb{C}^2$.

HW1 P7: repeated squaring over $\mathbb{F}_2$. **Idea.** Each squaring has two equal cross terms. In $\mathbb{F}_2$ they add to zero, so only the two squares remain. Computing powers $2, 4, 8$ uses this same calculation at each stage.

Here x is a formal variable; polynomial equality compares coefficients in the specified field.

1. First distribute all four terms:
$$(x+1)^2 = x^2 + x + x + 1 = x^2 + (1+1)x + 1 = x^2 + 1.$$
The last equality uses $1 + 1 = 0$ in $\mathbb{F}_2$.
2. Next square the result:
$$(x+1)^4 = (x^2+1)^2 = x^4 + x^2 + x^2 + 1 = x^4 + 1.$$
3. Square once more:
$$(x+1)^8 = (x^4+1)^2 = x^8 + x^4 + x^4 + 1 = x^8 + 1.$$
4. Over $\mathbb{R}$, the first squaring is already different: $(x+1)^2 = x^2 + 2x + 1$, with $2x$ retained. Repeated distribution over $\mathbb{R}$ gives
$$(x+1)^8 = x^8 + 8x^7 + 28x^6 + 56x^5 + 70x^4 + 56x^3 + 28x^2 + 8x + 1.$$
5. Every intermediate coefficient in this last expression is even. Reducing those integer coefficients modulo 2 gives zero for each, in agreement with $x^8 + 1$ over $\mathbb{F}_2$. Over $\mathbb{R}$ the seven coefficients remain nonzero.

Exam practice: Midterm 1 review, Section II, Exercise 7. **Idea.** For intersection, closure must be checked in each subspace using the same vectors. For union, choose one vector from each of two different lines; their sum may lie on neither line.

1. Let U, W be subspaces of V. Each contains 0_V, so $0_V \in U \cap W$.
2. For $u, v \in U \cap W$ and $a \in F$, $u, v \in U$ gives $au + v \in U$. Separately $u, v \in W$ gives $au + v \in W$. Therefore $au + v \in U \cap W$, and the subspace test proves the intersection claim (Notes Proposition 5).
3. For union, take $U = \{(t, 0) : t \in \mathbb{R}\}$ and $W = \{(0, t) : t \in \mathbb{R}\}$ in $\mathbb{R}^2$. They contain zero, and
$$a(t,0) + (s,0) = (at + s, 0) \in U, \qquad a(0,t) + (0,s) = (0, at + s) \in W.$$
Thus each is a subspace.
4. Yet $(1, 0)$ and $(0, 1)$ lie in their union, while $(1, 0) + (0, 1) = (1, 1)$ has neither coordinate zero. It belongs to neither U nor W. Thus the union need not be a subspace.